Experiment 15

Single-Level Directory Organizationcode:

Code:

```
#include <stdio.h>
#include <string.h>

int main() {
   char dir[10][20], file[10][10][20];
   int count[10] = {0};
   int nd = 0, choice, i, j, d;
   char dirname[20], filename[20];

   while (1) {
      printf("\n1. Create Directory");
      printf("\n2. Create File");
      printf("\n3. List Files");
      printf("\n4. Exit");
      printf("\nEnter choice: ");
      scanf("%d", &choice);

      switch (choice) {
      case 1:
         if (nd == 10) {
            printf("Directory limit reached\n");
            break;
         }

         printf("Enter directory name: ");
         scanf("%19s", dirname);
```

```
    for (i = 0; i < nd; i++) {
        if (strcmp(dir[i], dirname) == 0)
            break;
    }

    if (i == nd) {
        strcpy(dir[nd], dirname);
        count[nd] = 0;
        nd++;
        printf("Directory created successfully\n");
    } else {
        printf("Directory already exists\n");
    }
    break;

case 2:
    printf("Enter directory name: ");
    scanf("%19s", dirname);

    d = -1;
    for (i = 0; i < nd; i++)
        if (strcmp(dir[i], dirname) == 0)
            d = i;

    if (d == -1) {
        printf("Directory not found\n");
        break;
    }

    if (count[d] == 10) {
```

```
      printf("Directory is full\n");
      break;
    }

    printf("Enter file name: ");
    scanf("%19s", filename);

    found:
    for (j = 0; j < count[d]; j++) {
      if (strcmp(file[d][j], filename) == 0) {
        printf("File already exists\n");
        break;
      }
    }

    if (j == count[d]) {
      strcpy(file[d][count[d]], filename);
      count[d]++;
      printf("File created successfully\n");
    }
    break;

  case 3:
    for (i = 0; i < nd; i++) {
      printf("\nDirectory: %s\n", dir[i]);

      for (j = 0; j < count[i]; j++)
        printf("%s\n", file[i][j]);

      if (count[i] == 0)
        printf("No files\n");
```

```
        }
        break;

    case 4:
        return 0;

    default:
        printf("Invalid choice\n");
    }
  }

  return 0;
}
```

Output:

